Kshitiz

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.60
SAVITRI PUBLIC SCHOOL	Class XII (CBSE)	2023	85.80%
R P M ACADEMY	Class X (CBSE)	2021	88.20%

WORK EXPERIENCE

AI Developer Intern -- Allegient Unified Technology Pvt. Ltd.

- Researched and documented **enterprise AI use cases**, assessing each for **data readiness and feasibility**.
- Evaluated **Microsoft 365 Copilot** and **Copilot Studio** against internal workflow requirements.
- Authored **use-case specs** covering data sources, success metrics, and **permission-oversharing risks**.

PROJECTS

RAG Q&A System over ML Research Papers | GitHub | Live Demo

- **103,979 words** across **6 arXiv papers** (201 pages) indexed into a RAG pipeline with **FAISS**.
- **80%** precision@5 improvement **(0.114 0.206)** by tuning chunk size and overlap across **9** chunking configurations.
- **39**-question evaluation harness with source and page-span labels, plus **4 out-of-corpus** refusal controls.
- **436 chunks** retrieved by **cosine similarity** with page-span citations, deployed on **Streamlit**.

Abstractive Text Summarization: End-to-End Transformer | GitHub | Live Demo

- **11.68M parameter** encoder-decoder Transformer built from scratch in **PyTorch** with self and cross attention .
- **7 modules** hand-coded: masked self-attention, sinusoidal positional encoding, FFN, residuals and LayerNorm.
- **287K** CNN/DailyMail pairs trained with Adam, LR warmup, inverse-square-root decay, teacher forcing, dropout.
- **47% training loss reduction (5.76 to 3.04)** over **10 epochs**; unigram F1 **24.5%**, LCS F1 **19.9%** from scratch.
- **44% higher unigram F1, 117% higher LCS F1** vs TF-IDF extractive and BiLSTM + Bahdanau baselines.
- **80% consistency** Logistic Regression sentiment classifier; Random Forest topic model (30 trees, 5-fold CV).
- **10-page Streamlit dashboard** with real-time attention heatmaps and interactive NLP preprocessing explorer.
- **3 enterprise use cases** shown: knowledge management, automated reporting and media monitoring at scale.

JEE Intelligence Platform | GitHub | Live Demo

- **124,861** feature-engineered records across **23 IITs** (2016-2024) in an end-to-end **scikit-learn** ML pipeline.
- **5,240** institute-branch-category-gender combinations classified for IIT seat-allocation eligibility outcomes.
- **79.3% accuracy and 0.80 F1-score** with soft-voting ensemble (KNN, Decision Tree, Random Forest, AdaBoost).
- **82.6% accuracy (+3.3 pts)** MLP benchmarked; shipped the explainable ensemble after interpretability review.
- **5,349 combinations:** forecast next-year closing ranks via per-segment regression with confidence intervals.
- **9 years** of JEE Advanced data (2016-2024) cleaned and feature-engineered using **Python, Pandas and NumPy**.
- **5 classifiers** evaluated (KNN, Decision Tree, Random Forest, AdaBoost, MLP) with **5-fold cross-validation**.
- **23 IITs** of seat-allocation trends turned into data-driven insights guiding stakeholder decision making.

ACHIEVEMENTS

- **1000+ DSA problems** solved across coding platforms with 300+ active days in time-constrained environments.
- **Top 7%** of AtCoder competitors worldwide, demonstrating consistent algorithmic problem-solving skill.
- **3 independent** end-to-end technical projects spanning machine learning, deep learning/NLP and retrieval systems in 2 years.
- **3 live public tools** deployed for non-technical users with UX thinking and operational reliability.
- **60-70%** estimated analyst time saved per report by automating document reading and condensing.
- **4 evaluation frameworks** (ROUGE, Sentiment, Topic, retrieval precision@k) built for hypothesis-driven problem solving.

SKILLS

- **Programming Languages:** Python, C++
- **ML / Deep Learning:** Ensemble Learning (Decision Tree, Random Forest, KNN, AdaBoost), Neural Networks (MLP, BiLSTM, Transformer Architecture), Optimization (Adam, Gradient Clipping, Xavier Init, Dropout), Model Evaluation (Precision, Recall, F1, Cross-Validation, N-gram Overlap)
- **NLP:** Text Preprocessing (Tokenization, Stemming, Lemmatization, Stopword Removal), Feature Extraction (TF-IDF, Bag of Words, N-grams), Attention Mechanisms, Sequence-to-Sequence Modeling, Named Entity Recognition, Sentiment Analysis
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Plotly
- **Core Computer Science:** Data Structures & Algorithms , Object-Oriented Programming
- **Tools & Platforms:** Git, GitHub, scikit-learn, Streamlit, Joblib, Google Colab, Jupyter Notebook, Streamlit Community
- **Agile Practices:** Modular pipeline design, iterative benchmarking, version-controlled development using Git

RELEVANT COURSES

Elementary Linear Algebra (MA1140), Differential Equations (MA1150), Artificial Intelligence (ID1050), Computational Methods in Materials Science (MS2860)

CERTIFICATIONS

Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp| View Certificate

Decode C++ with DSA -- Physics Wallah, 2026| View Certificate

POSITIONS OF RESPONSIBILITY

Coordinator | Tinkerers' Lab, IIT Hyderabad

- Coordinated and managed technical workshops for 400+ students, ensuring smooth execution, participant engagement, and on-time delivery of sessions.
- Streamlined hostel accommodation logistics for event attendees, improving operational efficiency for large-scale campus events.
- Produced orientation reels for Tinkerers' Lab, showcasing lab facilities and activities to incoming students and driving awareness among the new batch.